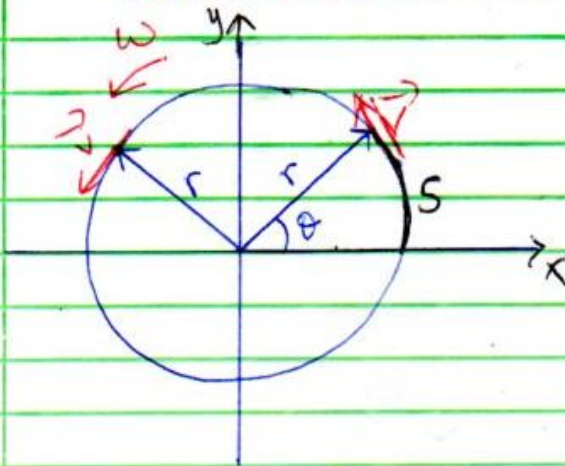


Topic 7 Summary: Uniform Circular Motion

- Definitions and units
 - Period and frequency
 - Angular velocity
 - Linear speed
 - Centripetal acceleration
 - Centripetal force
 - Radian measure
- Deriving equations for velocity and acceleration
- The conical pendulum
- Motion around a banked curve
 - Optimum banking angle/ design speed
 - No tendency to slip (without friction)
 - Tendency to slip outwards/ greatest speed
 - Tendency to slip inwards/ minimum speed
- Worked examples

UNIFORM CIRCULAR MOTION



→ speed of object is constant but velocity is changing.

Some definitions

T - period (time for one complete revolution)
unit: second

f - frequency (# revolutions per sec) $f = \frac{1}{T}$
unit: Hz

ω = angular velocity

$$\omega = \frac{2\pi}{T} = 2\pi f$$

unit: rad/s

$$\text{or } \omega = \frac{\theta}{t}$$

linear speed $v = \frac{2\pi r}{T} = \omega r$

ω +ve anti $\downarrow$
-ve clockwise

$$|a_c| = \frac{v^2}{r} = \omega^2 r$$

a_c is always directed toward centre of circle

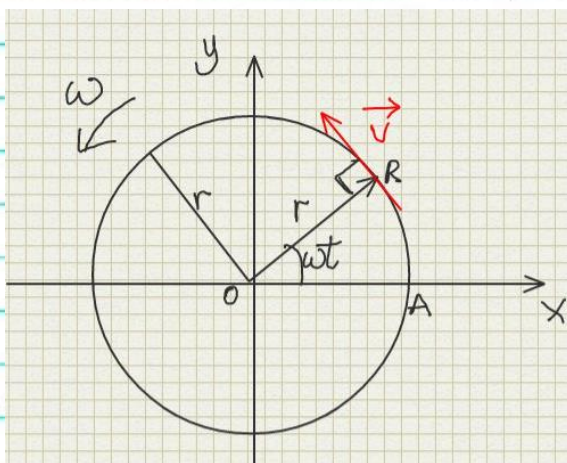
$\therefore$ centripetal acceleration.

$$s = r\theta$$

θ is in radians

Deriving Equations for $\vec{v}$ and $\vec{a}$

Consider a particle R moving around a circle, center O and radius, r . Let the angular velocity of R be ω rad/s.



Let the particle be at A ($r, 0$) when $t=0$ and at R, after t sec. Then $\angle ROA = \omega t$ since $\omega = \frac{\theta}{t}$.

The coordinates of R will be $(r \cos \omega t, r \sin \omega t)$.

The position vector of R in terms of the unit vectors $\hat{i}$ and $\hat{j}$ is

$$\begin{aligned}\vec{OR} = \vec{r} &= r \cos \omega t \hat{i} + r \sin \omega t \hat{j} \\ &= r [\cos \omega t \hat{i} + \sin \omega t \hat{j}]\end{aligned}$$

Differentiating $\vec{r}$ w.r.t t to obtain $\vec{v}$.

$$\begin{aligned}\vec{v} &= \frac{d\vec{r}}{dt} = r [-\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}] \\ &= r\omega [-\sin \omega t \hat{i} + \cos \omega t \hat{j}]\end{aligned}$$

and $\boxed{|\vec{v}| = r\omega}$

$\Rightarrow$ the velocity of R is of magnitude $r\omega$ in the direction of the unit vector $(-\sin \omega t \hat{i} + \cos \omega t \hat{j})$ which is $\perp$ to $\vec{OR}$.

Ex: Show that $\vec{r}$ and $\vec{v}$ are perpendicular.

Differentiating $\vec{v}$ w.r.t t to get $\vec{a}$.

$$\vec{a} = \frac{d\vec{v}}{dt} = r\omega [-\omega \cos \omega t \hat{i} - \omega \sin \omega t \hat{j}]$$

$$= -r\omega^2 [\cos \omega t \hat{i} + \sin \omega t \hat{j}]$$

$$= -\omega^2 [r \cos \omega t \hat{i} + r \sin \omega t \hat{j}]$$

$$\boxed{\vec{a} = -\omega^2 \vec{r}} \quad \text{--- (i)} \quad \rightarrow = \vec{r}$$

$$\therefore \boxed{|\vec{a}| = r\omega^2} \quad \text{--- (ii)}$$

From (ii), we see that the magnitude of the acceleration is $r\omega^2$, and from (i) we see that its direction is $-\vec{r}$ i.e. towards the center, O.

Example 1:

A vacuum cleaner has radius 10 cm and goes round at 600 rpm. What is its centripetal acceleration?

Sol.

$$f = \frac{600}{60} = 10 \text{ Hz} \quad \therefore T = \frac{1}{f} = \frac{1}{10} \text{ sec}$$

$$\omega = \frac{2\pi}{T} = 63 \text{ rad/s}$$

$$v = \omega r = 6.3 \text{ m/s}$$

$$a_c = \omega^2 r \approx 400 \text{ m/s}^2$$

Example 2:

Find the speed in m/s of a particle that is moving on a circular path of radius 30 cm with an angular speed of

- 4 rad/s
- 5 rev/min.

Example 2 : Sol.

a) $\omega = 4 \text{ rad/s}$ $30 \text{ cm} = 0.3 \text{ m}$
 $v = r\omega$
 $= 0.3 \times 4 = \underline{\underline{1.2 \text{ m/s}}}$

b) $\omega = 5 \text{ rev/min}$
 $= \frac{5 \times 2\pi}{60} \text{ rad/s}$
 $= \pi/6 \text{ rad/s}$
 $v = r\omega$
 $= 0.3 \times \pi/6 = \underline{\underline{\pi/20 \text{ m/s}}}$

Note: $\pi = 3.14$

Example 3:-

A small body rests on a horizontal turntable at a distance of 20 cm from the centre. Given that the body does not slide when the turntable is rotated at 33 rpm, find the maximum possible value of the coefficient of friction for the body and turntable.

Sol:



$$F = \mu mg = \frac{mv^2}{r} = mr\omega^2$$

$$33 \text{ rpm} = \frac{33 \times 2\pi}{60} = 3.456 \text{ rad/s}$$

Centripetal force = friction
Maximum possible value of friction is when body is at point of slipping

$$\mu mg = mr\omega^2$$

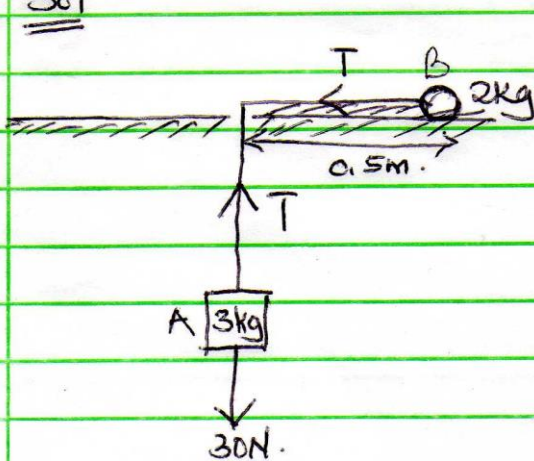
$$10\mu = 0.2 \times 3.456^2$$

$$\mu = \underline{\underline{0.2389}}$$

Example 4:

Particles A and B, with masses 3 kg and 2 kg respectively, are connected by a light inextensible string which passes through a hole in a table. A hangs vertically below the table while B describes a circle of radius 30 cm on the surface of the table which is smooth. Find the velocity of B if A is just held at rest.

Sol



centripetal force = Tension in string
 $T = 30\text{ N}$ (3kg mass)

$$\therefore \frac{mv^2}{r} = 30$$

$$\frac{2v^2}{0.5} = 30 \Rightarrow v^2 = 7.5$$

$$v = \underline{\underline{2.739\text{ m/s}}}$$

Force in Circular Motion (Centripetal Force)

By Newton's first law, if $\vec{v}$ is changing, this means that there must be a force acting. This force is given by $\vec{F} = m\vec{a}$ (Newton's 2nd law)

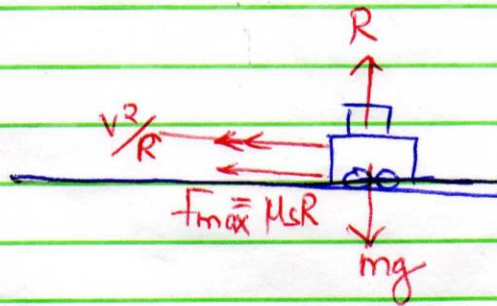
$$\vec{F} = m(-r\omega^2) = -mr\omega^2 = -\frac{mv^2}{r}$$

i.e. negative indicates $\vec{F}$ is towards centre of circle.

Example 5: Car on a circular Path.

A car travels on a horizontal road which is an arc of a circle of radius 125 m. The greatest speed at which the car can travel without slipping is 42 km/h. Find the coefficient of friction between the Tyres of the car, and the surface of the road.

Sol.



Let mass of car be m kg

Resolving vertically: $R = mg$ — (i)

Friction provides the force towards center of the circle and since car is at point of slipping,

$$F_{f \max} = \mu_s R = \frac{mv^2}{r} \quad \text{--- (ii)}$$

Eliminating R in (i) & (ii) gives

$$\mu_s mg = \frac{mv^2}{r}$$

$$\boxed{\mu_s = \frac{v^2}{rg}}$$

Given: $r = 125$ m, $v = \frac{42 \times 1000}{3600} = \frac{70}{6}$ m/s, $g = 9.8$

$$\mu_s = \left(\frac{70}{6}\right)^2 \times \frac{1}{125 \times 9.8} = \frac{1}{9}$$

The Conical Pendulum

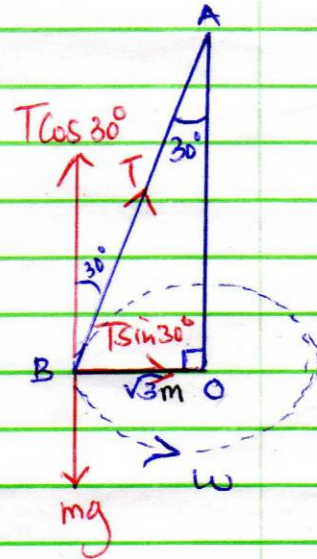
A particle is attached to the lower end of a light inextensible string, the upper end of which is fixed. When the particle describes a horizontal circle, the string describes the curved surface of a cone. This arrangement is known as a conical pendulum.

Example 6:

A conical pendulum consists of a light inextensible string AB, fixed at A and carrying a particle of mass 50g at B.

The particle moves in a horizontal circle of radius $\sqrt{3}\text{m}$ and center vertically below A.

If the angle between the string and the vertical is 30° , find the tension in the string and the angular speed of the particle. ($g = 9.8\text{m/s}^2$)



Sol:

Resolving vertically:

$$T \cos 30^\circ = \frac{50}{1000} g$$

$$T = 0.566 \text{ N}$$

The horizontal component of the tension provides the centripetal force.

i.e.

$$T \sin 30^\circ = m r \omega^2$$

$$0.283 = 0.05 \times \sqrt{3} \omega^2$$

$$\omega^2 = 3.268$$

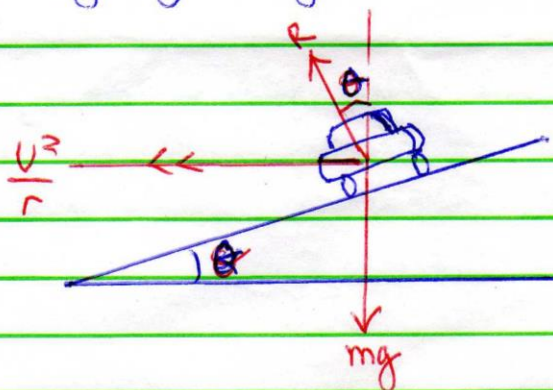
$$\omega = 1.81 \text{ rad/s.}$$

Motion of a car around a banked Curve:

When a car travels along a circular arc on a horizontal road, the frictional force between the Tyres and the road provides the necessary horizontal force F towards the centre of the circular arc.

If the speed of the car exceeds a particular value then the force required ($\frac{mv^2}{r}$) may be greater than can be provided by the max. frictional force, and the car will slip.

In practice, a bend on a road or race track may be banked so that the edge at the outside of the bend is slightly higher than that at the inside.



The normal reaction R between the car and road is no longer vertical. The horizontal component of R can therefore provide or help to provide the necessary centripetal force.

The direction of the frictional force will be up or down the slope depending on whether the car has a tendency to slip sideways towards inside or outside of bend.

Case I: Car has no tendency to slip

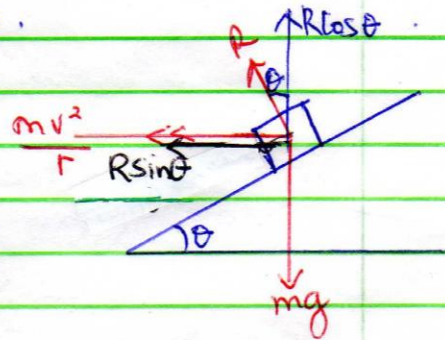
(10)

Example 7:

A car travels around a bend of radius 400m on a road which is banked at an angle θ to the horizontal. If the car has no tendency to slip when travelling at 35 m/s, find θ .

Sol:

Since no tendency to slip, there is no frictional force between tyres and road.



Vertically: $R \cos \theta = mg$ ---- (i)

Horizontally: $R \sin \theta = \frac{mv^2}{r}$ ---- (ii)

Divide (ii) by (i)

$$\frac{R \sin \theta}{R \cos \theta} = \frac{\frac{mv^2}{r}}{mg}$$

$$\tan \theta = \frac{v^2}{rg}$$

Given $r = 400 \text{ m}$, $v = 35 \text{ m/s}$

$$\tan \theta = \frac{35^2}{400 \times 9.81} = 0.312$$

$$\theta = \underline{\underline{17.34^\circ}}$$

Example 8:

A car is rounding a bend of radius 100m which is banked at angle of 10° to the horizontal. At what speed must the car travel to ensure there is no tendency to slip sideways.

Sol:

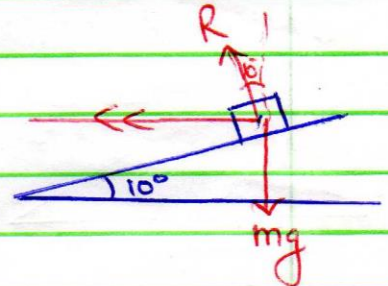
Vertically: $R \cos 10^\circ = mg$ — (i)

Horizontally: $R \sin 10^\circ = \frac{mv^2}{r}$

$$\text{i.e. } \tan 10^\circ = \frac{v^2}{rg}$$

$$\begin{aligned} v^2 &= rg \tan 10^\circ \\ &= 100 \times 9.81 \times \tan 10^\circ \\ &= 172.98 \end{aligned}$$

$$v = 13.15 \text{ m/s}$$



Case II: Car travelling at high speed, tendency to slip outwards.

Example 9:

A car travels around a bend in a road which is a circular arc of radius 62.5 m. The road is banked at an angle $\tan^{-1} \frac{5}{12}$ to the horizontal. If the coefficient of friction between the tyres and the road is 0.4, find

- the greatest speed at which the car can be driven round the bend without slipping.
- the least speed at which this can happen.

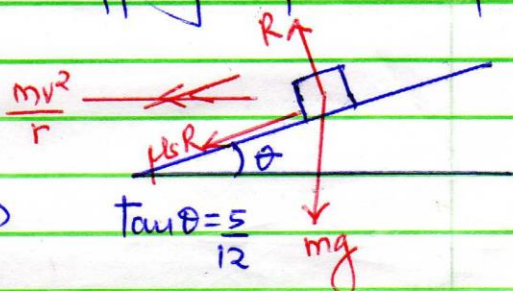
Sol:

- a) When the car is travelling as fast as possible, the max frictional force acts μR acts down the slope so as to prevent the car slipping up the slope.

Sol:

Vertically:

$$R \cos \theta = mg + \mu R \sin \theta \quad \text{--- (i)}$$



Horizontally:

$$\frac{mv^2}{r} = \mu R \cos \theta + R \sin \theta \quad \text{--- (ii)}$$

$$\therefore \sin \theta = \frac{5}{13} \quad \cos \theta = \frac{12}{13}$$

Eliminating R gives

$$R(\cos \theta - \mu \sin \theta) = mg$$

$$R(\mu \cos \theta + \sin \theta) = \frac{mv^2}{r}$$

$$\text{i.e. } \frac{mg}{\cos \theta - \mu \sin \theta} = \frac{mv^2}{r(\mu \cos \theta + \sin \theta)}$$

$$\frac{rg}{v^2} = \frac{\cos \theta - \mu \sin \theta}{\mu \cos \theta + \sin \theta}$$

$$\text{Given } r = 62.5, \mu = 0.4, \sin \theta = 5/13, \cos \theta = 12/13$$

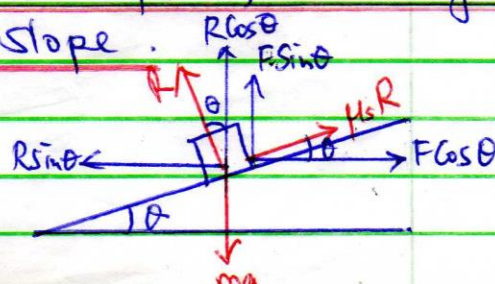
$$v^2 = 600.86$$

$$v = 24.5 \text{ m/s}$$

Case III: Car travelling at low speed, tendency to slip down the slope.

b) Vertically:

$$R \cos \theta + \mu R \sin \theta = mg \quad \text{--- (i)}$$



Horizontally:

$$R \sin \theta - \mu R \cos \theta = \frac{mv^2}{r} \quad \text{--- (ii)}$$

$$\text{for (i)} \quad \frac{13}{13} R + \frac{0.4 \times 5}{13} R = 9.81 \text{ m}$$

$$\frac{14}{13} R = 9.81 \text{ m} \quad \Rightarrow \quad \frac{R}{\text{m}} = 9.81 \times \frac{13}{14} = 9.12$$

$$\text{(ii)} \quad \frac{5}{13} R - \frac{0.4 \times 12}{13} R = \frac{mv^2}{62.5}$$

$$0.015 R = \frac{mv^2}{62.5}$$

$$v^2 = \frac{0.96 R}{\text{m}}$$

$$= 8.7589 \quad v = \underline{\underline{2.96 \text{ m/s}}}$$